Code Appendix

Academic Field

```r
library(ggplot2)

salary <- read.table("salary.txt", header = TRUE)

salary$field <- as.factor(salary$field)
salary$sex <- as.factor(salary$sex)

field_palette <- c("#1b9e77", "#d95f02", "#7570b3", "#e7298a", "#66a61e")

base_theme <- theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "right"
  )
```

```r
cat("\nACADEMIC FIELD - ALL YEARS\n")
```

```
##
## ACADEMIC FIELD - ALL YEARS
```

```r
field_counts_all <- table(salary$field)
print(field_counts_all)
```

```
##
##  Arts Other  Prof
##  2840 13143  3809
```

```r
field_props_all <- prop.table(field_counts_all)
print(round(field_props_all, 3))
```

```
##
##  Arts Other  Prof
## 0.143 0.664 0.192
```

```r
field_sex_counts_all <- table(salary$field, salary$sex)
print(field_sex_counts_all)
```

```
##
##             F     M
##   Arts    802  2038
##   Other  2735 10408
##   Prof    389  3420
```

```r
field_sex_props_all <- prop.table(field_sex_counts_all, margin = 2)
print(round(field_sex_props_all, 3))
```

```
##
```

```
##             F     M
##   Arts  0.204 0.128
##   Other 0.697 0.656
##   Prof  0.099 0.216
```

```r
print(chisq.test(field_sex_counts_all))
```

```
##
##  Pearson's Chi-squared test
##
## data:  field_sex_counts_all
## X-squared = 355.79, df = 2, p-value < 2.2e-16
```

```r
p1 <- ggplot(salary, aes(x = field)) +
  geom_bar(fill = field_palette[1]) +
  labs(
    title = "Overall Distribution of Academic Field (All Years)",
    x = "Academic Field",
    y = "Count"
  ) +
  base_theme

ggsave("overall_field_counts_all_years.png", p1, width = 8, height = 6)

p2 <- ggplot(salary, aes(x = field)) +
  geom_bar(aes(y = after_stat(count / sum(count))), fill = field_palette[2]) +
  labs(
    title = "Proportion of Faculty by Academic Field (All Years)",
    x = "Academic Field",
    y = "Proportion"
  ) +
  base_theme

ggsave("overall_field_proportion_all_years.png", p2, width = 8, height = 6)

p3 <- ggplot(salary, aes(x = sex, fill = field)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = field_palette) +
  labs(
    title = "Distribution of Academic Field by Sex (All Years)",
    x = "Sex",
    y = "Proportion",
    fill = "Field"
  ) +
  base_theme

ggsave("stacked_field_by_sex_all_years.png", p3, width = 8, height = 6)
```

```r
data95 <- subset(salary, year == 95)

cat("\nACADEMIC FIELD - 1995 ONLY\n")
```

```
##
## ACADEMIC FIELD - 1995 ONLY
```

```r
field_counts_95 <- table(data95$field)
print(field_counts_95)
```

```
##
##  Arts Other  Prof
##   220  1067   310
```

```r
field_props_95 <- prop.table(field_counts_95)
print(round(field_props_95, 3))
```

```
##
##  Arts Other  Prof
## 0.138 0.668 0.194
```

```r
field_sex_counts_95 <- table(data95$field, data95$sex)
print(field_sex_counts_95)
```

```
##
##           F   M
##   Arts   80 140
##   Other 287 780
##   Prof   42 268
```

```r
field_sex_props_95 <- prop.table(field_sex_counts_95, margin = 2)
print(round(field_sex_props_95, 3))
```

```
##
##             F     M
##   Arts  0.196 0.118
##   Other 0.702 0.657
##   Prof  0.103 0.226
```

```r
print(chisq.test(field_sex_counts_95))
```

```
##
##  Pearson's Chi-squared test
##
## data:  field_sex_counts_95
## X-squared = 37.955, df = 2, p-value = 5.73e-09
```

```r
p4 <- ggplot(data95, aes(x = field)) +
  geom_bar(fill = field_palette[1]) +
  labs(
    title = "Overall Distribution of Academic Field (1995)",
    x = "Academic Field",
    y = "Count"
  ) +
  base_theme

ggsave("overall_field_counts_1995.png", p4, width = 8, height = 6)

p5 <- ggplot(data95, aes(x = sex, fill = field)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = field_palette) +
  labs(
    title = "Distribution of Academic Field by Sex (1995)",
    x = "Sex",
```

```r
    y = "Proportion",
    fill = "Field"
  ) +
  base_theme

ggsave("stacked_field_by_sex_1995.png", p5, width = 8, height = 6)

png("mosaic_field_by_sex_all_years.png", width = 800, height = 600)
mosaicplot(field ~ sex, data = salary,
           main = "Mosaic Plot of Academic Field by Sex (All Years)",
           col = field_palette)
dev.off()
```

```
## pdf
##   2
```

```r
png("mosaic_field_by_sex_1995.png", width = 800, height = 600)
mosaicplot(field ~ sex, data = data95,
           main = "Mosaic Plot of Academic Field by Sex (1995)",
           col = field_palette)
dev.off()
```

```
## pdf
##   2
```

```r
png("difference_in_proportions_all_years.png", width = 800, height = 600)

diff_props <- field_sex_props_all[, "F"] - field_sex_props_all[, "M"]

barplot(diff_props,
        main = "Difference in Proportion (Female - Male)\nAll Years",
        ylab = "Difference in Proportion",
        col = field_palette[3])

dev.off()
```

```
## pdf
##   2
```

Year Attained Highest Degree

```r
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)

col_names_11 <- c(
  "case","id","sex","deg","yrdeg","field",
  "startyr","year","rank","admin","salary"
)

df <- read.table(
  "salary.txt",
  header = TRUE,
  col.names = col_names_11,
  fill = TRUE,
  strip.white = TRUE,
  stringsAsFactors = FALSE
)

str(df)
```

```
## 'data.frame':    19792 obs. of  11 variables:
##  $ case   : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ id     : int  1 2 2 4 6 6 6 6 6 7 ...
##  $ sex    : chr  "F" "M" "M" "M" ...
##  $ deg    : chr  "Other" "Other" "Other" "PhD" ...
##  $ yrdeg  : int  92 91 91 96 66 66 66 66 66 70 ...
##  $ field  : chr  "Other" "Other" "Other" "Other" ...
##  $ startyr: int  95 94 94 95 91 91 91 91 91 71 ...
##  $ year   : int  95 94 95 95 91 92 93 94 95 76 ...
##  $ rank   : chr  "Assist" "Assist" "Assist" "Assist" ...
##  $ admin  : int  0 0 0 0 1 1 0 0 0 0 ...
##  $ salary : num  6684 4743 4881 4231 11182 ...
```

```r
head(df)
```

```
##   case id sex   deg yrdeg field startyr year   rank admin salary
## 1    1  1   F Other    92 Other      95   95 Assist     0   6684
## 2    2  2   M Other    91 Other      94   94 Assist     0   4743
## 3    3  2   M Other    91 Other      94   95 Assist     0   4881
## 4    4  4   M   PhD    96 Other      95   95 Assist     0   4231
## 5    5  6   M   PhD    66 Other      91   91   Full     1  11182
## 6    6  6   M   PhD    66 Other      91   92   Full     1  11507
```

```r
print(min(df$yrdeg, na.rm = TRUE))
```

```
## [1] 48
```

```r
print(max(df$yrdeg, na.rm = TRUE))
```

```
## [1] 96
```

```r
df <- df %>%
  mutate(yrdeg = 1900 + yrdeg)

head(df)
```

```
##   case id sex   deg yrdeg field startyr year   rank admin salary
## 1    1  1   F Other  1992 Other      95   95 Assist     0   6684
## 2    2  2   M Other  1991 Other      94   94 Assist     0   4743
```

```
## 3    3  2   M Other  1991 Other      94   95 Assist     0   4881
## 4    4  4   M   PhD  1996 Other      95   95 Assist     0   4231
## 5    5  6   M   PhD  1966 Other      91   91   Full     1  11182
## 6    6  6   M   PhD  1966 Other      91   92   Full     1  11507
```

```
df1 <- df %>%
  arrange(id, year) %>%
  group_by(id) %>%
  summarise(
    sex   = first(sex),
    yrdeg = first(yrdeg),
    .groups = "drop"
  )

head(df1)
```

```
## # A tibble: 6 x 3
##      id sex   yrdeg
##   <int> <chr> <dbl>
## 1     1 F      1992
## 2     2 M      1991
## 3     4 M      1996
## 4     6 M      1966
## 5     7 M      1970
## 6     8 M      1975
```

```
colSums(is.na(df1))
```

```
##    id   sex yrdeg
##     0     0     0
```

```
male   <- df1 %>% filter(sex == "M") %>% pull(yrdeg)
female <- df1 %>% filter(sex == "F") %>% pull(yrdeg)

cat("Male n:", length(male), "\n")
```

```
## Male n: 1188
```

```
cat("Female n:", length(female), "\n")
```

```
## Female n: 409
```

```
male_mean   <- mean(male)
female_mean <- mean(female)

male_median   <- median(male)
female_median <- median(female)

cat("Male mean yrdeg:", male_mean, "\n")
```

```
## Male mean yrdeg: 1974.369
```

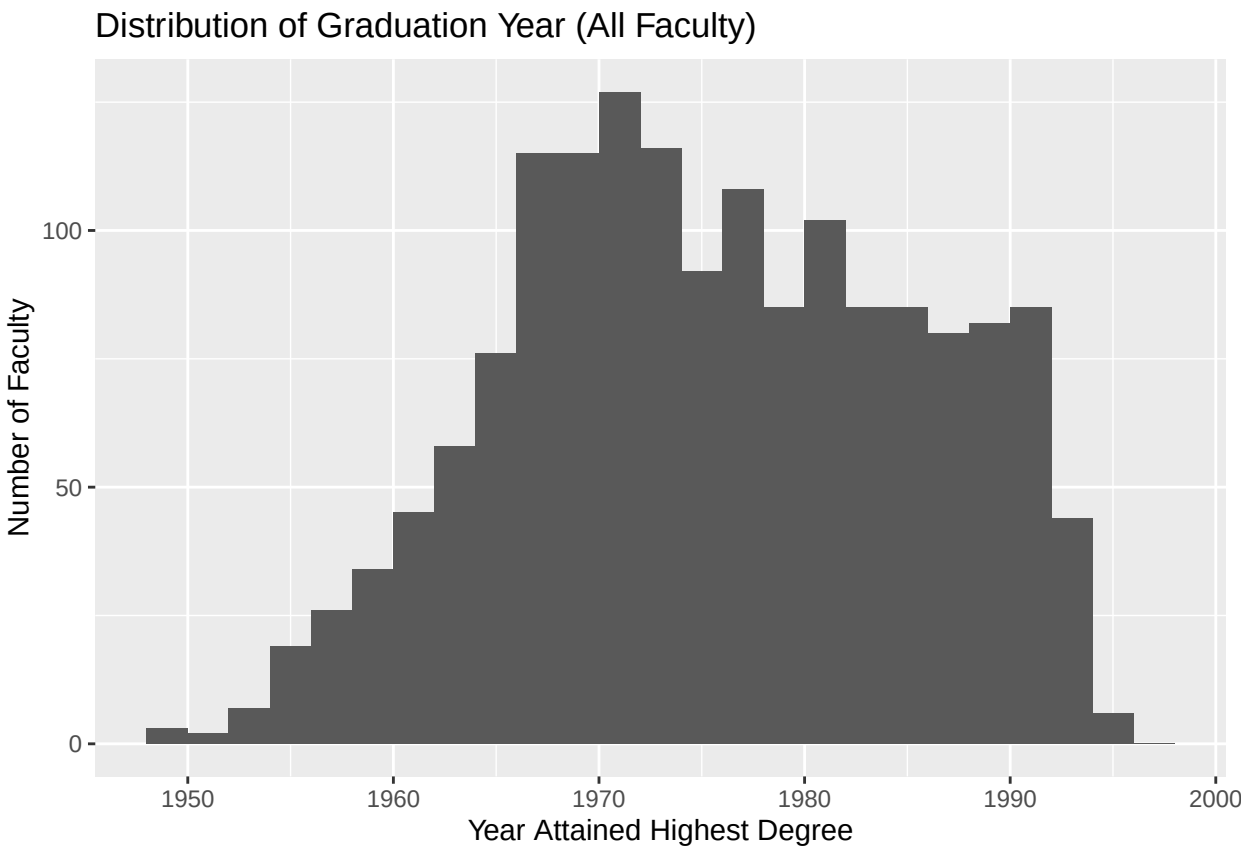
```
cat("Female mean yrdeg:", female_mean, "\n")
```

```
## Female mean yrdeg: 1981.108
```

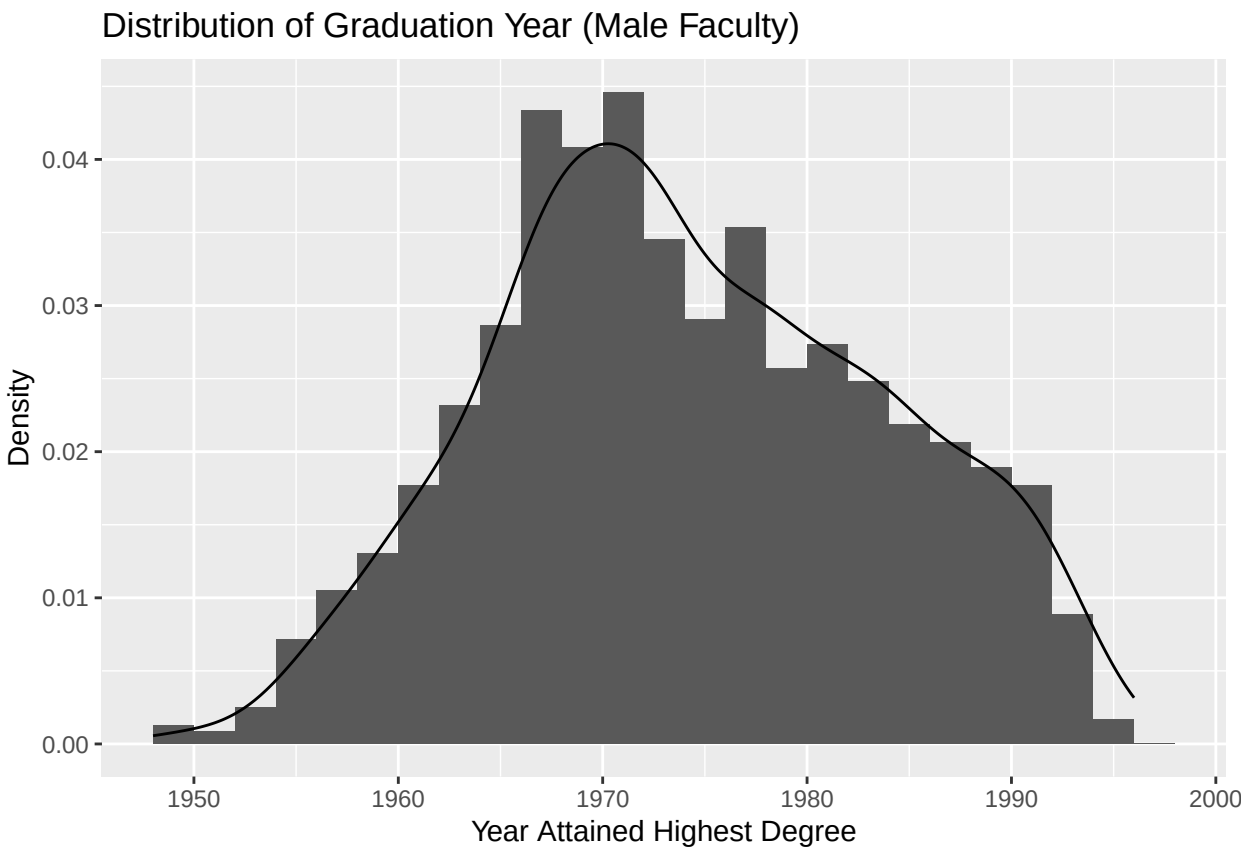
```
cat("Difference in means (M - F):", male_mean - female_mean, "\n")
```

```
## Difference in means (M - F): -6.738893
```

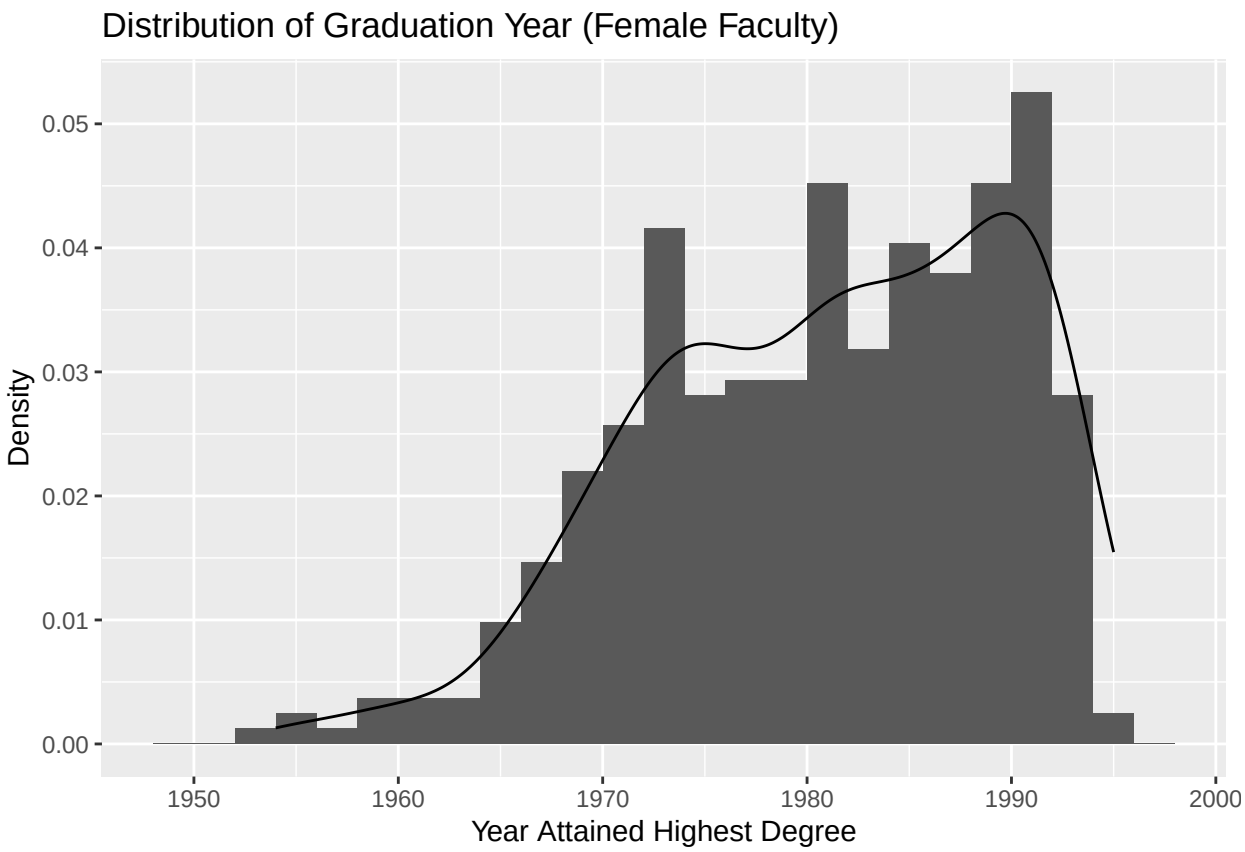
```r
bins <- seq(min(df1$yrdeg), max(df1$yrdeg) + 2, by = 2)

ggplot(df1, aes(x = yrdeg)) +
  geom_histogram(breaks = bins) +
  labs(
    x = "Year Attained Highest Degree",
    y = "Number of Faculty",
    title = "Distribution of Graduation Year (All Faculty)"
  )
```




```r
ggplot(data.frame(yrdeg = male), aes(x = yrdeg)) +
  geom_histogram(aes(y = after_stat(density)), breaks = bins) +
  geom_density() +
  labs(
    x = "Year Attained Highest Degree",
    y = "Density",
    title = "Distribution of Graduation Year (Male Faculty)"
  )
```



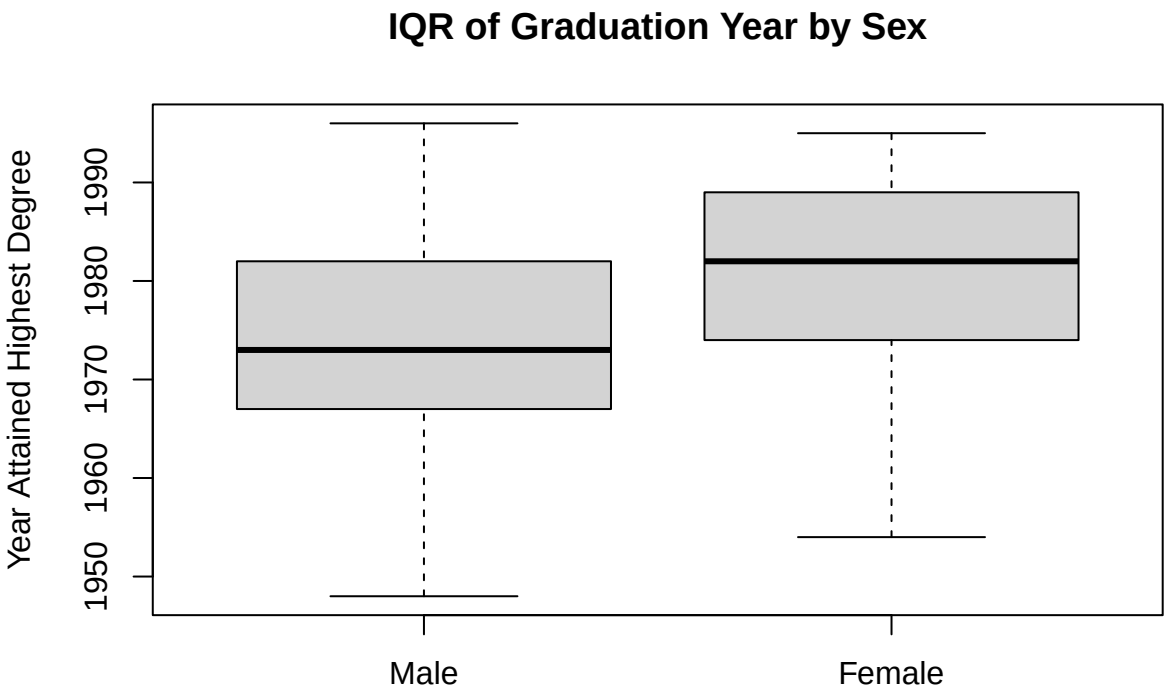

```r
ggplot(data.frame(yrdeg = female), aes(x = yrdeg)) +
  geom_histogram(aes(y = after_stat(density)), breaks = bins) +
  geom_density() +
  labs(
    x = "Year Attained Highest Degree",
    y = "Density",
    title = "Distribution of Graduation Year (Female Faculty)"
  )
```




```r
df1 %>%
  group_by(sex) %>%
  summarise(
    count  = n(),
    mean   = mean(yrdeg),
    median = median(yrdeg),
    std    = sd(yrdeg),
    .groups = "drop"
  )
```

```
## # A tibble: 2 x 5
##   sex   count  mean median   std
##   <chr> <int> <dbl>  <dbl> <dbl>
## 1 F       409 1981.   1982  8.70
## 2 M      1188 1974.   1973  9.64
```

```r
boxplot(
  male, female,
  names = c("Male", "Female"),
  ylab = "Year Attained Highest Degree",
  main = "IQR of Graduation Year by Sex"
)
```




```r
tt <- t.test(male, female, var.equal = FALSE)

cat("t-statistic:", unname(tt$statistic), "\n")
```

```
## t-statistic: -13.13177
```

```r
cat("p-value:", tt$p.value, "\n")
```

```
## p-value: 1.002651e-35
```

Year Hired at the University

```r
df <- read.table("salary.txt", header = TRUE)

head(df)
```

```
##   case id sex   deg yrdeg field startyr year   rank admin salary
## 1    1  1   F Other    92 Other      95   95 Assist     0   6684
## 2    2  2   M Other    91 Other      94   94 Assist     0   4743
## 3    3  2   M Other    91 Other      94   95 Assist     0   4881
## 4    4  4   M   PhD    96 Other      95   95 Assist     0   4231
## 5    5  6   M   PhD    66 Other      91   91   Full     1  11182
## 6    6  6   M   PhD    66 Other      91   92   Full     1  11507
```

```r
df$year_hired <- 1900 + df$startyr

head(df[, c("sex", "startyr", "year_hired")])
```

```
##   sex startyr year_hired
## 1   F      95       1995
## 2   M      94       1994
## 3   M      94       1994
## 4   M      95       1995
```

```
## 5   M      91       1991
## 6   M      91       1991
```

```r
df %>%
  group_by(sex) %>%
  summarise(
    count = n(),
    mean  = mean(year_hired, na.rm = TRUE),
    std   = sd(year_hired, na.rm = TRUE)
  )
```

```
## # A tibble: 2 x 4
##   sex   count  mean   std
##   <chr> <int> <dbl> <dbl>
## 1 F      3926 1980.  8.09
## 2 M     15866 1975.  8.83
```

```r
c(count = length(df$year_hired), mean = mean(df$year_hired, na.rm = TRUE), std = sd(df$year_hired, na.r
```

```
##        count         mean          std
## 19792.000000  1976.072201     8.951341
```

```r
cat("Overall\n")
```

```
## Overall
```

```r
print(summary(df$year_hired))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1948    1969    1976    1976    1983    1995
```

```r
cat("\nBy sex\n")
```

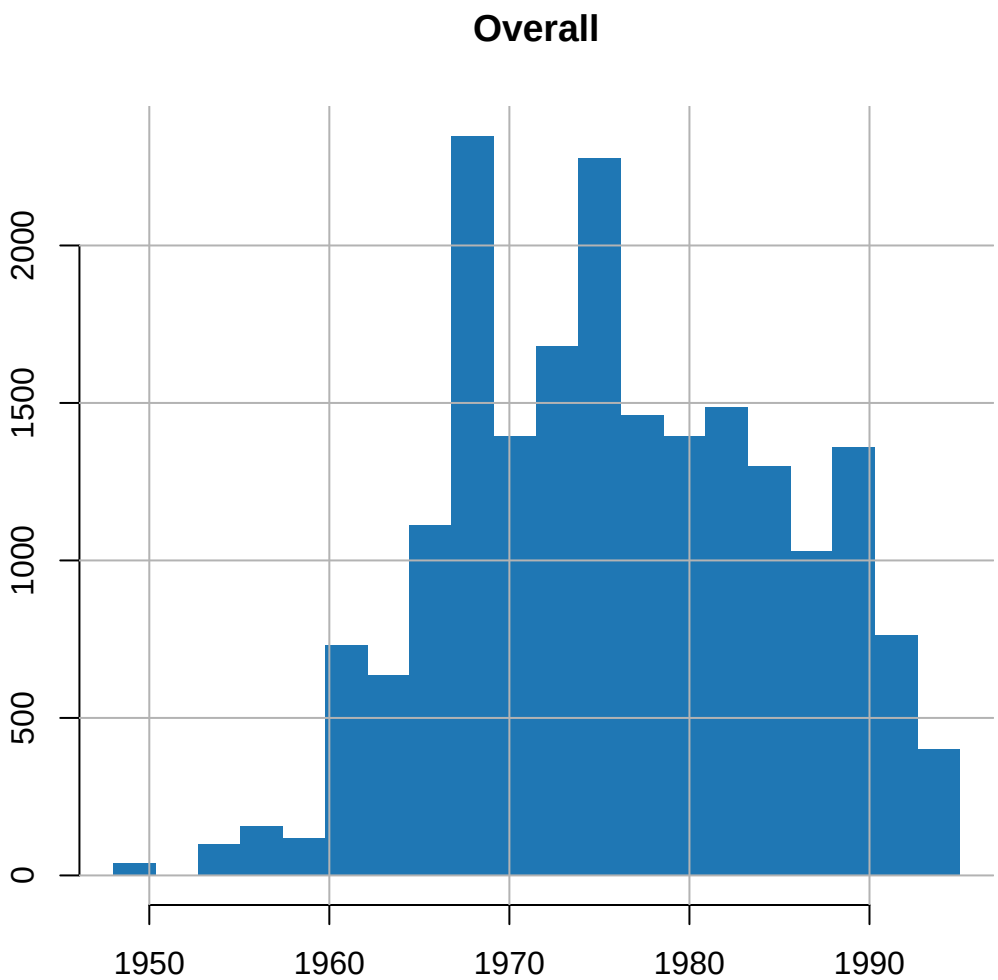
```
##
## By sex
```

```r
print(by(df$year_hired, df$sex, summary))
```

```
## df$sex: F
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1957    1974    1980    1980    1987    1995
## ------------------------------------------------------------
## df$sex: M
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1948    1968    1974    1975    1982    1995
```

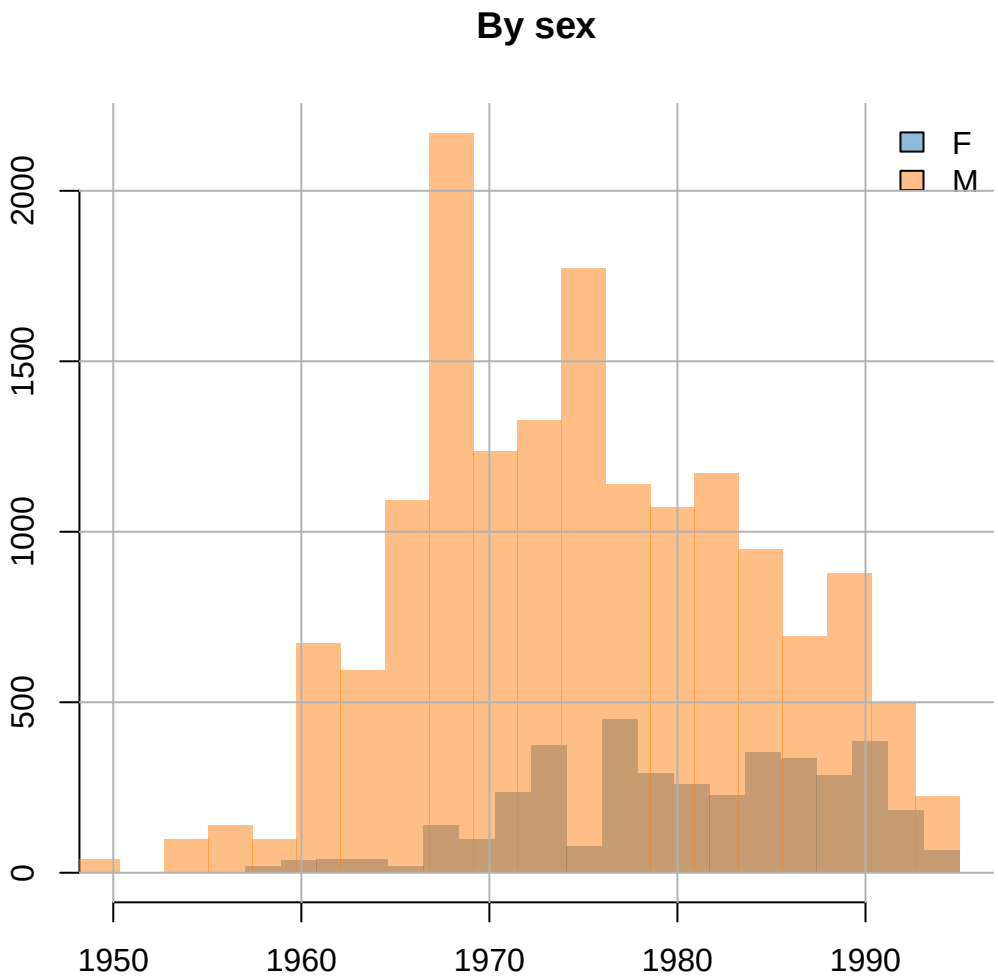
```r
rng <- range(df$year_hired, na.rm = TRUE)
breaks <- seq(rng[1], rng[2], length.out = 21)

blue   <- "#1f77b4"
orange <- "#ff7f0e"

add_mpl_grid <- function() {
  usr <- par("usr")
  x_ticks <- axTicks(1)
  y_ticks <- axTicks(2)
  abline(v = x_ticks, col = "grey70", lwd = 1)
  abline(h = y_ticks, col = "grey70", lwd = 1)
}
```

```r
hist(df$year_hired,
     breaks = breaks,
     right = FALSE,
     col = blue,
     border = NA,
     main = "Overall",
     xlab = "",
     ylab = "")
add_mpl_grid()
```

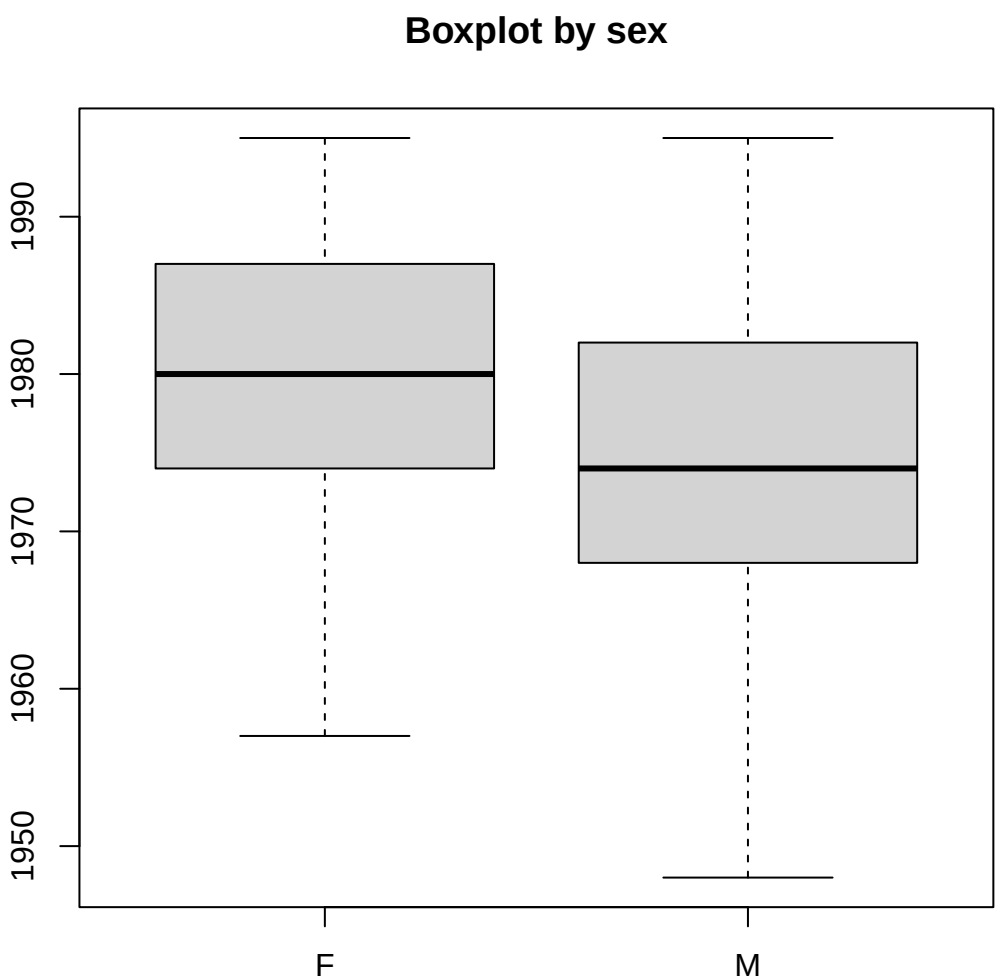



```r
hF <- hist(df$year_hired[df$sex == "F"], breaks = breaks, right = FALSE, plot = FALSE)
hM <- hist(df$year_hired[df$sex == "M"], breaks = breaks, right = FALSE, plot = FALSE)
ylim_shared <- c(0, max(c(hF$counts, hM$counts)))

breaksF <- seq(min(df$year_hired[df$sex=="F"]), max(df$year_hired[df$sex=="F"]), length.out=21)
breaksM <- seq(min(df$year_hired[df$sex=="M"]), max(df$year_hired[df$sex=="M"]), length.out=21)

hF <- hist(df$year_hired[df$sex=="F"], breaks=breaksF, right=FALSE, plot=FALSE)
hM <- hist(df$year_hired[df$sex=="M"], breaks=breaksM, right=FALSE, plot=FALSE)
ylim_shared <- c(0, max(c(hF$counts, hM$counts)))

hist(df$year_hired[df$sex=="F"], breaks=breaksF, right=FALSE,
```

```r
     col=adjustcolor("#1f77b4", 0.5), border=NA, ylim=ylim_shared,
     xlim=c(1950, max(df$year_hired, na.rm=TRUE)),
     main="By sex", xlab="", ylab="")

hist(df$year_hired[df$sex=="M"], breaks=breaksM, right=FALSE,
     col=adjustcolor("#ff7f0e", 0.5), border=NA, add=TRUE)

legend("topright", legend=c("F","M"),
       fill=c(adjustcolor("#1f77b4",0.5), adjustcolor("#ff7f0e",0.5)),
       bty="n")
add_mpl_grid()
```




```r
boxplot(year_hired ~ sex,
        data = df,
        main = "Boxplot by sex",
        xlab = "",
        ylab = "")
```



Academic Rank in 1995

```
df <- read.table("salary.txt", header = TRUE)
df95 <- subset(df, year == 95)
df95$rank <- factor(df95$rank, levels = c("Assist", "Assoc", "Full"))
df95$sex <- factor(df95$sex)

rank_counts <- table(df95$rank)
rank_props <- prop.table(rank_counts)

data.frame(
  rank = names(rank_counts),
  count = as.integer(rank_counts),
  proportion = as.numeric(rank_props)
)
```

```
##     rank count proportion
## 1 Assist   315  0.1972448
## 2  Assoc   437  0.2736381
## 3   Full   845  0.5291171
```

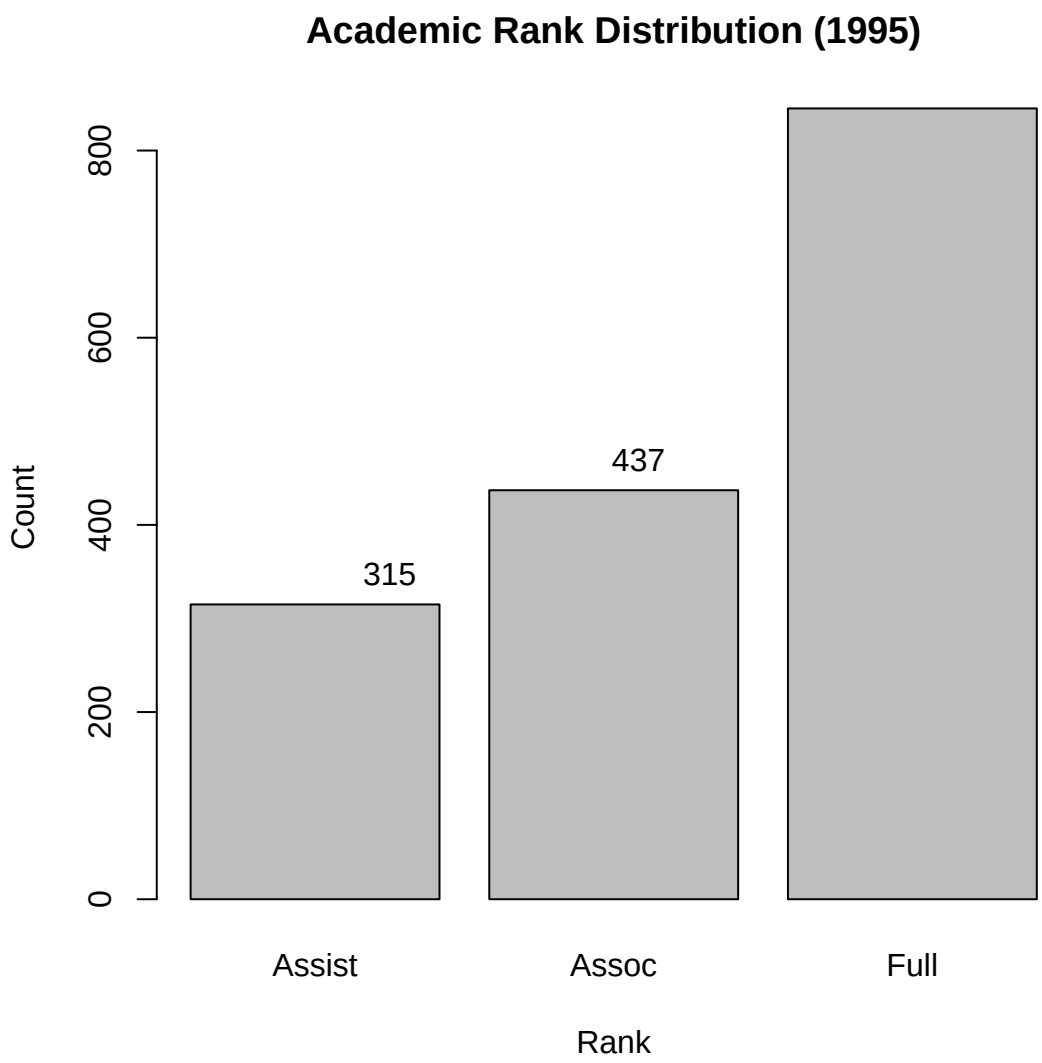
```r
rank_by_sex <- table(df95$sex, df95$rank)
rank_by_sex
```

```
##
##     Assist Assoc Full
##   F    145   138  126
##   M    170   299  719
```

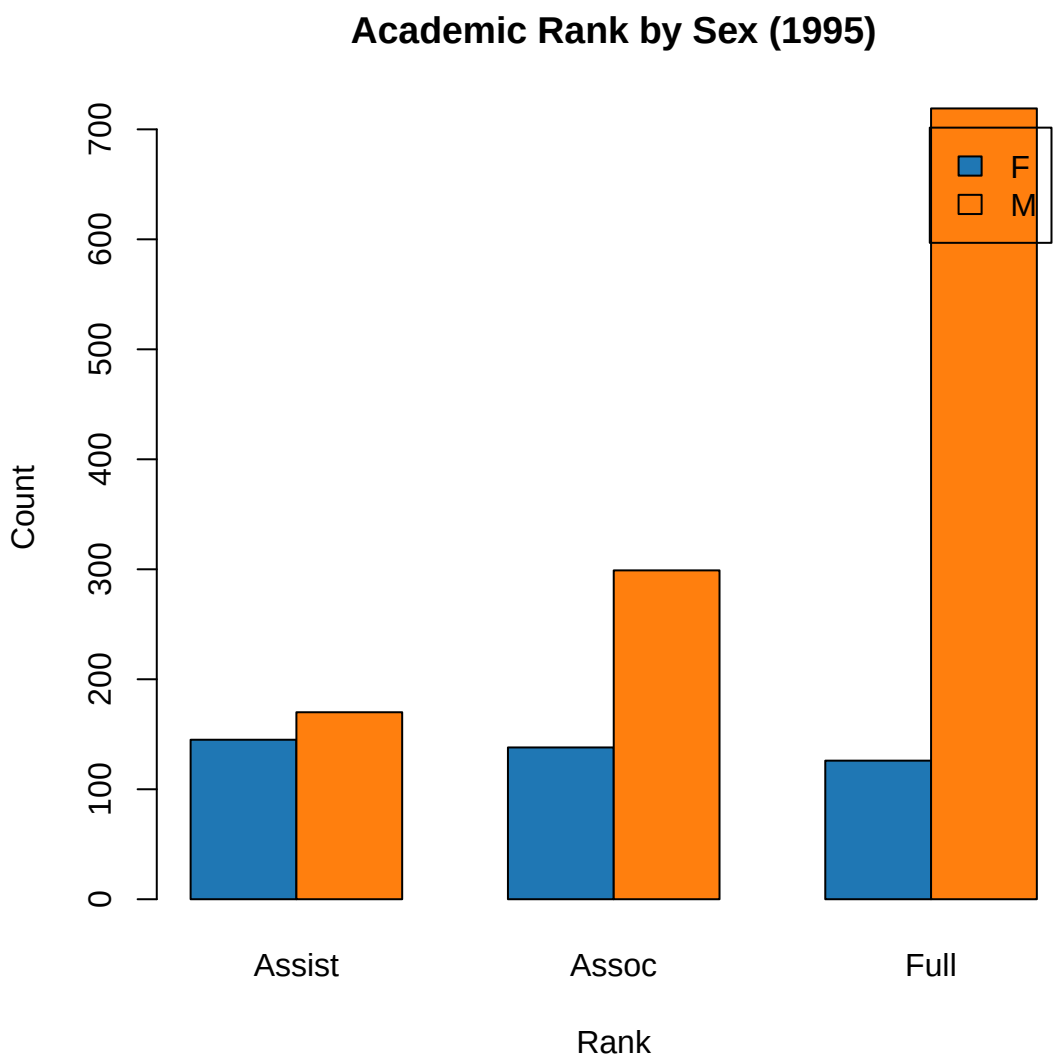
```r
rank_by_sex_prop <- prop.table(rank_by_sex, margin = 1)
round(rank_by_sex_prop, 3)
```

```
##
##     Assist Assoc  Full
##   F  0.355 0.337 0.308
##   M  0.143 0.252 0.605
```

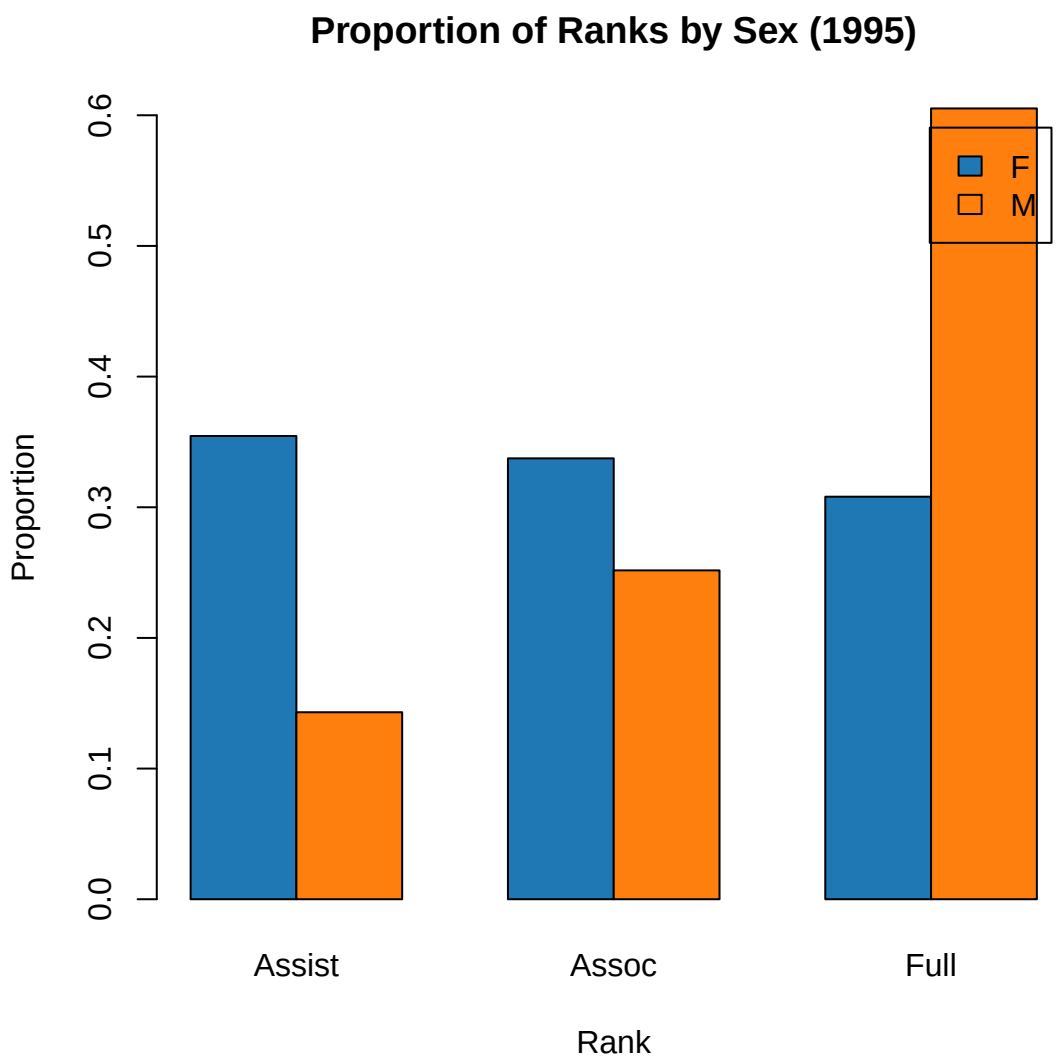
```r
barplot(rank_counts,
        main = "Academic Rank Distribution (1995)",
        xlab = "Rank",
        ylab = "Count")

text(seq_along(rank_counts),
     rank_counts,
     labels = rank_counts,
     pos = 3)
```




```r
barplot(rank_by_sex,
        beside = TRUE,
        col = c("#1f77b4", "#ff7f0e"),
        main = "Academic Rank by Sex (1995)",
        xlab = "Rank",
        ylab = "Count",
        legend.text = rownames(rank_by_sex))
```




```r
barplot(rank_by_sex_prop,
        beside = TRUE,
        col = c("#1f77b4", "#ff7f0e"),
        main = "Proportion of Ranks by Sex (1995)",
        xlab = "Rank",
        ylab = "Proportion",
        legend.text = rownames(rank_by_sex_prop))
```



Monthly Salary in 1995

```
df <- read.table("salary.txt", header = TRUE, sep = "")
head(df)
```

```
##   case id sex   deg yrdeg field startyr year   rank admin salary
## 1    1  1   F Other    92 Other      95   95 Assist     0   6684
## 2    2  2   M Other    91 Other      94   94 Assist     0   4743
## 3    3  2   M Other    91 Other      94   95 Assist     0   4881
## 4    4  4   M   PhD    96 Other      95   95 Assist     0   4231
## 5    5  6   M   PhD    66 Other      91   91   Full     1  11182
## 6    6  6   M   PhD    66 Other      91   92   Full     1  11507
```

```
missing_counts_by_group <- df %>%
  group_by(sex) %>%
  summarise(
    Missing_Value = sum(is.na(salary))
    )

# Print the result
```

```r
print(missing_counts_by_group)
```

```
## # A tibble: 2 x 2
##   sex   Missing_Value
##   <chr>         <int>
## 1 F                 0
## 2 M                 0
```

```r
mean_salary <- mean(df$salary, na.rm = TRUE)
median_salary <- median(df$salary, na.rm = TRUE)
mean_salary
```

```
## [1] 4721.823
```

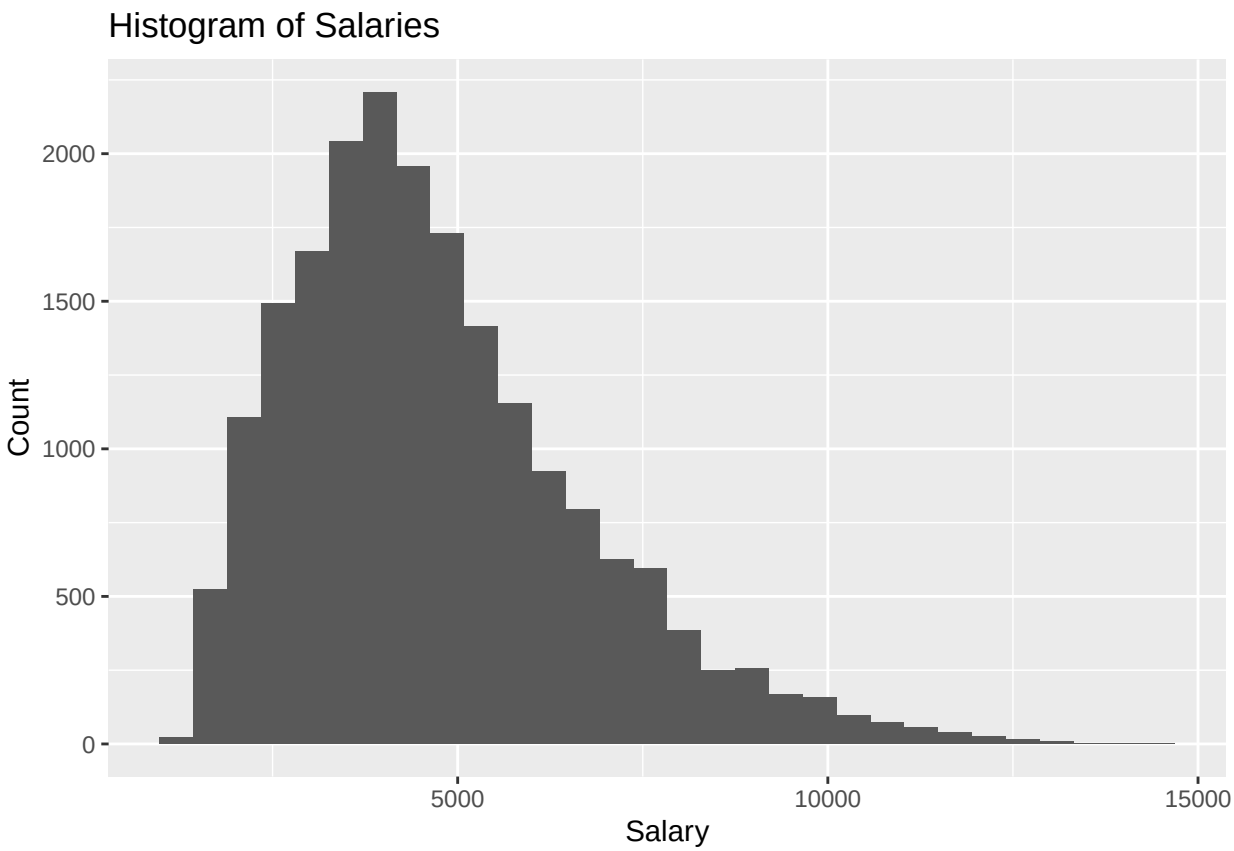
```r
median_salary
```

```
## [1] 4353
```

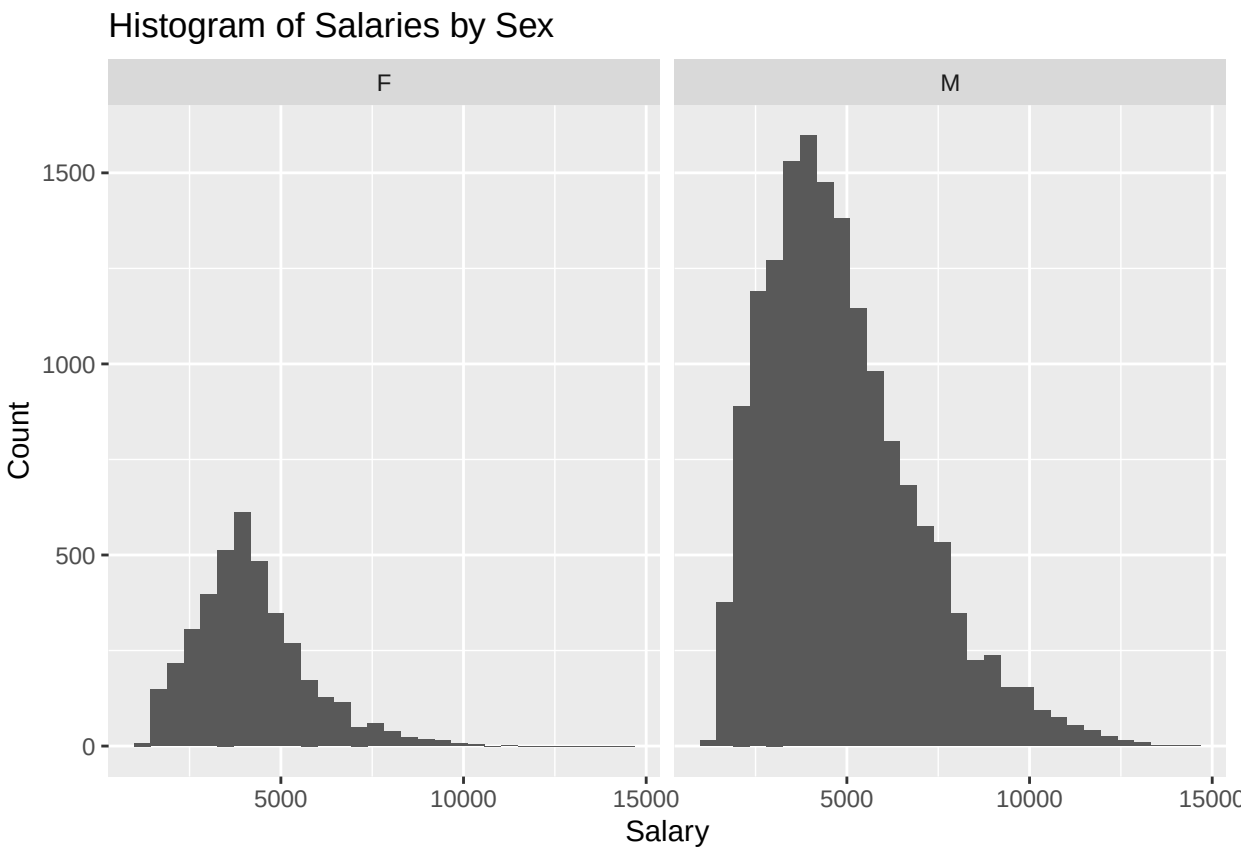
```r
df %>%
  group_by(sex) %>%
  summarise(
    mean_salary = mean(salary, na.rm = TRUE),
    median_salary = median(salary, na.rm = TRUE)
    )
```

```
## # A tibble: 2 x 3
##   sex   mean_salary median_salary
##   <chr>       <dbl>         <dbl>
## 1 F           4185.          3996
## 2 M           4855.          4505
```

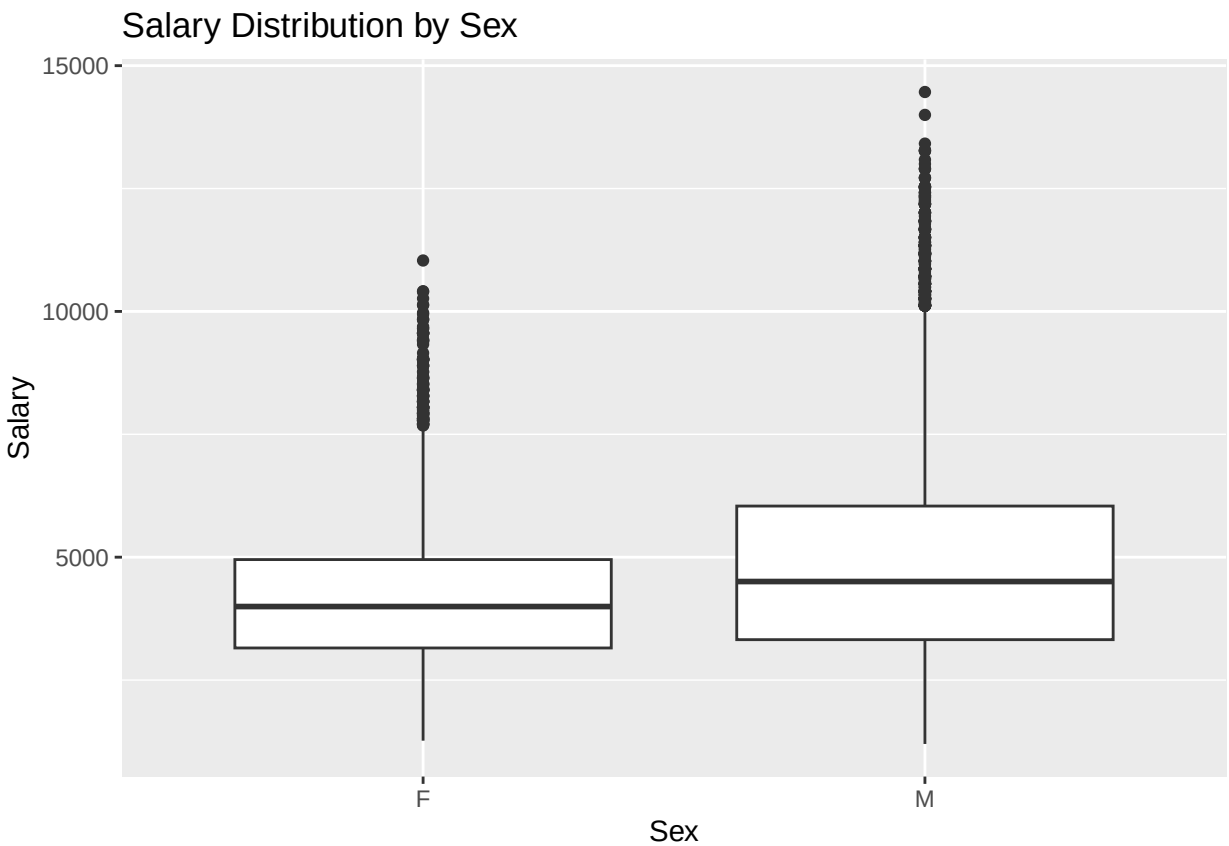
```r
ggplot(df, aes(x = salary)) +
  geom_histogram(bins = 30) +
  theme_gray() +
  labs(
    title = "Histogram of Salaries",
    x = "Salary",
    y = "Count"
    )
```



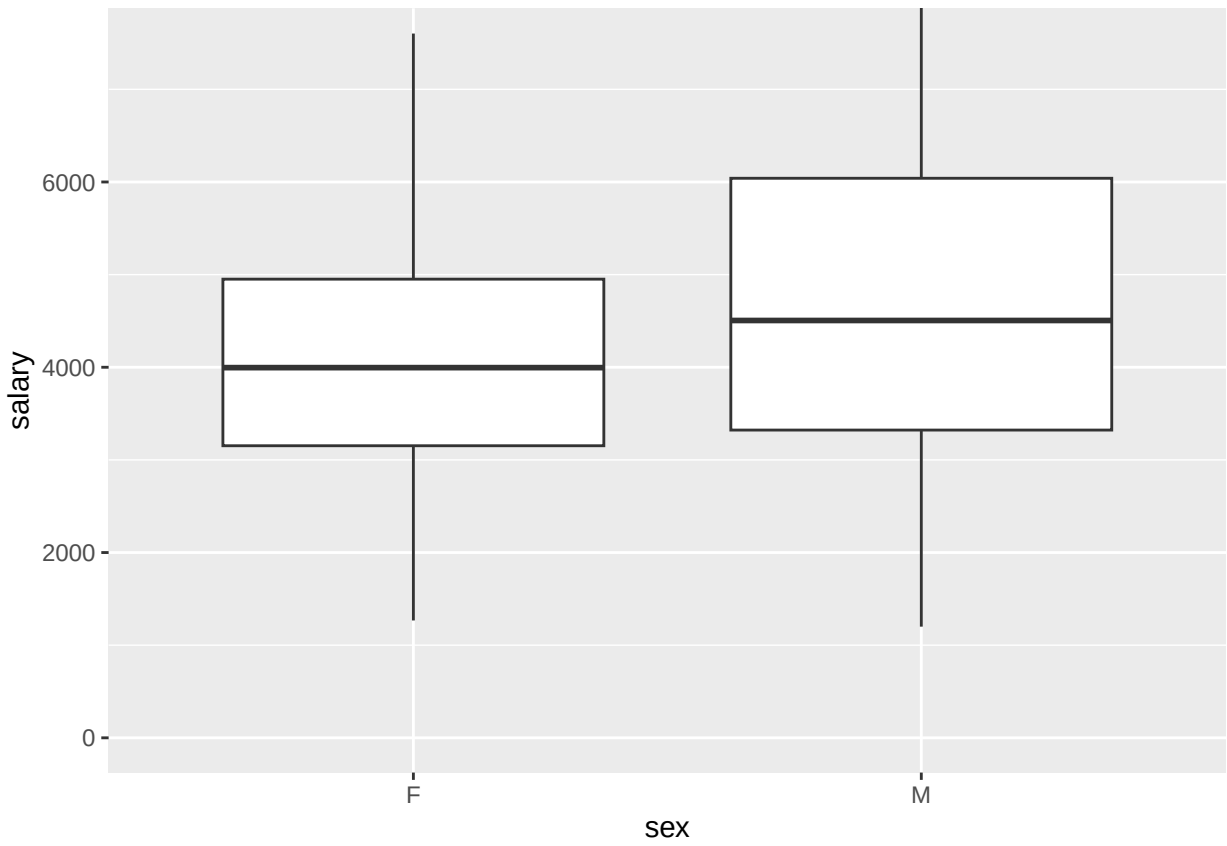

```r
ggplot(df, aes(x = salary)) +
  geom_histogram(bins = 30) +
  facet_wrap(~ sex) +
  theme_gray()+
  labs(
    title = "Histogram of Salaries by Sex",
    x = "Salary",
    y = "Count"
    )
```




```r
ggplot(df, aes(x = sex, y = salary)) +
  geom_boxplot() +
  theme_grey() +
  labs(
    title = "Salary Distribution by Sex",
    x = "Sex",
    y = "Salary"
    )
```




```r
ggplot(df, aes(x = sex, y = salary)) +
  geom_boxplot(outlier.shape = NA) +
  coord_cartesian(ylim = c(0, 7500)) # Zoom in to see the boxes clearly
```


6000
4000
2000
0
salary
F
M
sex